

Fee Payment Management Platform

Team-wise Work Assignment Document

Project Title

Fee Payment Management Platform

Problem Statement

Design and develop a secure REST-based platform to manage student fee payments, payment records, receipts, and financial reports. The platform should provide authenticated and authorized access using REST APIs, HTTPS, and JWT/OAuth-based security. It should support concurrent payment processing and secure logging to ensure data protection, reliability, and traceability.

Common Responsibilities (All Teams)

Before beginning development, all teams must jointly finalize:

- Software Requirement Specification (SRS)
- ER Diagram
- Database Schema
- Table Names and Primary Keys
- Coding Standards
- Folder Structure
- Naming Conventions
- REST API Design
- API Documentation (Swagger/OpenAPI)
- Git Repository Structure (if applicable)
- HTTPS Configuration Strategy
- JWT/OAuth Authentication Flow
- Integration Plan

- Test Plan

TEAM 1

Worklet: Student, Authentication & Fee Management

Objective

Develop the core platform responsible for student management, authentication, fee structure management, and fee payment initiation.

Major Modules

- Student Registration
- Login Authentication
- JWT/OAuth Authentication
- Role-Based Access Control
- Student Profile Management
- Fee Structure Management
- Fee Payment Initiation
- Fee Status Tracking
- Payment History

Database Tables

- Students
- Users
- Roles
- FeeStructure
- FeePayments

Subject-wise Implementation

Subject	Expected Implementation
Basic Programming	Modular program design, input validation, exception handling

Object-Oriented Programming	Student, User, Fee, Payment classes
Data Structures	Arrays/Lists for payment history and student records
DBMS	CRUD operations, SQL queries, constraints, foreign keys
Computer Networks	REST API endpoints, HTTPS communication
Operating Systems	Authentication logs and secure file handling
Software Engineering	SRS, UML, Use Case Diagram, Class Diagram
Web Technology	Login Page, Student Dashboard, Fee Payment Screens

Deliverables

- Login Module
- JWT Authentication Module
- Student Management Module
- Fee Structure Module
- Fee Payment Initiation Module
- REST API Documentation
- Database Scripts
- UML Diagrams
- Unit Testing Report

TEAM 2

Worklet: Payment Processing & Transaction Management

Objective

Develop secure payment processing, transaction management, receipt generation, and concurrent payment handling.

Major Modules

- Payment Processing

- Online Payment Gateway Simulation
- Payment Verification
- Payment Status Updates
- Receipt Generation
- Transaction History
- Payment Rollback
- Concurrent Payment Handling

Database Tables

- Transactions
- Receipts
- PaymentGatewayLogs

Subject-wise Implementation

Subject	Expected Implementation
Basic Programming	Payment processing logic and validation
Object-Oriented Programming	Transaction, Receipt, PaymentGateway classes
Data Structures	Queue for payment request processing
DBMS	Transaction management, ACID properties, SQL transactions
Computer Networks	Secure REST APIs, HTTPS communication
Operating Systems	Thread synchronization for concurrent payment requests
Software Engineering	Activity Diagram, Sequence Diagram
Web Technology	Payment Interface, Receipt Pages

Deliverables

- Payment Processing Module
- Transaction Management Module
- Receipt Generation Module
- Concurrency Demonstration
- REST API Documentation

- Database Scripts
- UML Diagrams
- Unit Testing Report

TEAM 3

Worklet: Reports, Security Monitoring & System Administration

Objective

Develop financial reporting, audit logging, security monitoring, administrative features, and integrate all project modules.

Major Modules

- Fee Collection Reports
- Pending Fee Reports
- Student Payment Reports
- Dashboard & Analytics
- Audit Logs
- Secure Log Management
- Backup & Restore
- System Monitoring
- Admin Dashboard

Database Tables

- Reports
- AuditLogs
- SecurityLogs
- BackupHistory

Subject-wise Implementation

Subject	Expected Implementation
Basic Programming	Report generation and log processing
Object-Oriented Programming	Report, LogManager, SecurityMonitor classes
Data Structures	Queue/Stack for log management
DBMS	Aggregate queries, report generation
Computer Networks	Secure REST API access using HTTPS
Operating Systems	Secure logging, backup scheduling, file management
Software Engineering	Integration Testing, Documentation
Web Technology	Dashboard, Reports, Charts, Export Pages

Deliverables

- Admin Dashboard
- Financial Reports Module
- Audit Log Module
- Security Monitoring Module
- Backup & Restore Module
- Integration Report
- Testing Report

Key Performance Indicators (KPIs)

Sr. No.	Key Performance Indicator (KPI)	Target
1	Functional Completeness	100% implementation of all modules as per the problem statement
2	Payment Automation	100% automation of fee payment, receipt generation, and reporting

3	Authentication & Authorization	Secure JWT/OAuth authentication with 100% role-based access
4	REST API Reliability	All REST APIs function correctly with proper HTTP status codes and validation
5	Secure Communication	100% HTTPS-enabled communication between client and server
6	Concurrent Payment Handling	Support at least 10 simultaneous payment requests without race conditions or data inconsistency
7	Transaction Accuracy	100% accurate fee payment and receipt generation
8	Report Generation	Financial reports generated within ≤ 5 seconds
9	Code Quality & Documentation	Complete SRS, UML, ER Diagram, API Documentation, Database Schema, and Testing Reports
10	System Integration & Reliability	Successful integration of all three worklets with no critical runtime errors during final demonstration

Expected Final Outcome

At the end of the semester, the integrated platform should support:

- Secure Login using JWT/OAuth
- Role-Based Access Control
- Student Profile Management
- Fee Structure Management
- Online Fee Payment
- Secure REST APIs
- HTTPS Communication
- Payment Processing
- Receipt Generation
- Concurrent Payment Processing
- Financial Reports
- Dashboard & Analytics
- Audit & Security Logs
- Backup & Restore
- Database-Driven Operations
- Complete API Documentation
- Complete Documentation and Testing